

Give CFG for following languages.

① $L = a^*b^*$

→ a^*b^* means any number of a's (including 0) followed by any number of b's (including 0)

note: all a's must come before any b's.

so, $L = \{\epsilon, a, aa, aaa, b, bb, ab, abb, aabbb, aaabbb, \dots\}$

→ for a^* : repeat 'a' any number of times or stop (ϵ).

$$\text{so, } A \rightarrow aA \mid \epsilon$$

→ for b^* : same logic as for a^*

$$\text{so, } B \rightarrow bB \mid \epsilon$$

→ Now combine both: $S \rightarrow AB$

so, final CFG can be:

$$S \rightarrow AB$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

② $L = \{a^{n+2}b^n \mid n \geq 0\}$

$$n=0 \rightarrow a^{0+2}b^0 = a^2 = aa = a^2$$

$$n=1 \rightarrow a^{1+2}b^1 = a^3b = aaab = a^2a^1b^1$$

$$n=2 \rightarrow a^{2+2}b^2 = a^4b^2 = aaaabb = a^2a^2b^2$$

$\vdots$

$$n=n \rightarrow a^{n+2}b^n = \underline{a^2 \cdot a^n b^n}$$

} → first 2 fixed a's
Then generate
equal no of a's
and b's.

→ CFG for $a^n b^n$ is: $S \rightarrow aSb \mid \epsilon$